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Ex-1 Process Creation and PID Display using System Calls in C.

AIM:-

To create a new process using system calls and display the Process ID (PID) and Parent Process ID (PPID) using a C Program.

Algorithm:-

1. Include necessary header files like stdio.h and unistd.h.
2. Declare variables to store process ID and parent process ID
3. Use getpid() to get the current process ID.
4. Use getppid() to get the parent process ID.
5. Print the process ID and parent process ID.
6. Use fork() system call to create a new process
7. Check the return value of fork():
   - If 0 → child process
   - If >0 → Parent process
8. Display process IDs for both parent and child processes

O/P verified 8/4/26

Result:-

The program successfully created a new process using the fork() system call and displayed the Process ID and Parent Process ID for both parent and child processes.

2. File copy using system calls in C

Aim:-

To identify and use system calls to copy the contents of one file to another using a C program

Algorithm

1. Include required header files (stdio.h, fcntl.h, unistd.h).
2. Open source file in read mode and destination file in write mode
3. Read data from sources file using read () or fgetc ()
4. Write data to destination file using write () or fputc ()
5. Close both files using close () or fclose ()

O/P verified

RESULT:-

The program successfully copied the contents from the source file to the destination file.

3. CPU scheduling using First come First served (FCFS)

AIM:-

To design a CPU scheduling program in C using the First come First served technique.

Algorithm:-

1. Input no. of processes along with arrival time and burst time.
2. Arrange processes based on arrival time (FCFS order)
3. Calculate waiting time for each process
4. calculate turn around time for each process.
5. Compute average waiting time and turn around time

o/p verified

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RESULT:-

The FCFS scheduling algorithm was implemented successfully and the waiting time and turnaround time for all processes were calculated.

4. Construct a scheduling Program using C to select the waiting process with SJF.

AIM :

To write a C program that implements the SJF CPU scheduling algorithm which select the process with the smallest burst time for execution.

Algorithm :-

1. Start the program and include necessary header files like stdio.h.

2. Read the number of processes and their burst time values from the user.

3. Arrange the processes in ascending order of burst time.

4. Calculate the waiting time and turnaround time for each process.

5. Display the processes details and compute the average waiting time and average turnaround time

O/P Verified

Validated

RESULT :-

Thus the C Program to implement SJF scheduling algorithm was executed successfully and the waiting time and turnaround time were calculated.

5. Construct a scheduling program using C to select the waiting process with the highest priority.

Aim:-

To write a C program that implements the Priority Scheduling algorithm where the process with the highest priority is executed first.

Algorithm:-

1. Start the program and include necessary header files like stdio.h

2. Read the number of processes along with the burst time and priority for each process.

3. Arrange the processes according to priority order

4. Calculate the waiting time and turn around time for each process.

5. Display the process details and compute average waiting time and average turn around time

O/P verified

Result:-

Thus the C program to implement Priority Scheduling algorithm was executed successfully and the waiting time and turnaround time were obtained.